

WAUTOMA, WI, USA

WMO: 720398

Lat: 44.042N

Lon: 89.304W

Elev: 859

StdP: 14.25

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.2	(c) -7.5	(d) -18.2	(e) 2.1	(f) -9.2	(g) -14.6	(h) 2.6	(i) -6.5	(j) 24.4	(k) 14.1	(l) 20.9	(m) 13.8	(n) 4.3	(o) 290	(p) 0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 23.7	(c) 90.4	(d) 72.5	(e) 87.6	(f) 71.7	(g) 84.0	(h) 70.0	(i) 76.0	(j) 85.4	(k) 74.3	(l) 83.1	(m) 72.6	(n) 80.7	(o) 9.5	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	127.0	80.0	71.9	121.9	78.7	70.1	114.5	76.9	40.1	86.3	38.3	82.9	36.7	81.1	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	18.3	16.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-20.9	79.3	5.3	1.5	-24.7	80.4	-27.9	81.3	-30.9	82.2	-34.7	83.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	15.8	19.6	32.0	44.4	57.0	66.5	71.0	68.5	60.8	48.1	34.8	22.2
	DBStd	21.25	12.29	11.52	11.92	9.03	8.92	6.27	5.61	5.54	7.76	8.75	9.97	10.83
	HDD50	4211	1061	853	571	216	31	0	0	0	8	146	464	861
	HDD65	7749	1526	1272	1024	620	280	55	12	28	171	529	906	1326
	CDD50	2454	0	0	13	48	246	496	650	574	332	87	8	0
	CDD65	518	0	0	0	1	31	101	198	137	45	5	0	0
	CDH74	5772	0	0	11	44	486	1187	2126	1304	529	85	0	0
	CDH80	1863	0	0	0	6	161	387	757	385	151	16	0	0
Wind	WSAvg	6.3	7.1	6.7	6.2	8.0	7.3	5.9	5.2	4.8	5.1	6.2	6.7	6.7
Precipitation	PrecAvg	32.2	1.3	1.1	1.8	3.1	4.0	4.7	3.9	3.9	3.1	2.4	1.8	1.4
	PrecMax	38.4	2.9	2.0	3.8	5.6	7.6	8.8	9.0	9.6	8.1	5.4	4.7	3.0
	PrecMin	26.2	0.5	0.5	0.3	1.1	1.9	1.7	1.3	1.3	0.6	0.4	0.0	0.3
	PrecStd	3.3	0.6	0.4	1.0	1.1	1.5	2.3	2.2	1.8	1.7	1.3	1.3	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.0	56.4	76.7	80.6	90.8	93.1	99.4	91.1	89.6	81.9	69.3	54.7
		MCWB	39.4	47.1	61.5	60.7	71.2	73.4	75.2	72.6	73.7	67.4	56.7	52.9
	2%	DB	39.5	44.9	64.4	72.8	84.1	88.7	90.5	88.0	84.2	74.8	61.9	45.1
		MCWB	35.8	39.9	54.2	57.2	66.5	71.8	73.0	72.4	71.0	63.6	53.7	41.7
	5%	DB	36.5	39.1	57.1	67.3	79.6	84.0	86.8	84.0	80.6	69.8	55.5	41.0
		MCWB	34.4	35.5	47.9	52.8	63.3	68.9	71.7	70.9	67.2	60.4	48.3	38.8
	10%	DB	34.1	35.8	50.0	62.8	74.2	81.0	83.8	81.3	75.4	64.0	50.3	36.9
		MCWB	32.8	32.6	43.2	50.7	60.9	66.8	70.4	69.2	65.3	57.1	45.2	34.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.1	48.4	63.8	64.5	73.1	77.4	78.4	77.4	75.6	71.4	59.1	51.6
		MCDB	44.6	57.1	75.8	73.4	87.6	87.0	89.6	86.0	86.0	79.3	66.0	52.6
	2%	WB	37.0	40.8	57.7	59.5	69.5	74.4	76.2	75.2	72.9	65.8	54.7	42.5
		MCDB	39.6	45.2	64.0	69.6	79.9	83.9	86.2	83.1	81.0	70.6	61.3	43.6
	5%	WB	34.7	36.1	49.4	55.6	66.8	72.0	74.5	73.1	70.4	62.0	49.8	38.5
		MCDB	36.0	39.2	56.8	64.7	75.0	80.3	83.5	80.1	76.0	67.5	55.6	40.6
	10%	WB	32.4	32.5	43.8	51.3	63.5	69.4	72.7	71.5	67.7	57.7	45.5	35.0
		MCDB	34.1	35.1	49.8	60.8	70.6	76.3	81.0	78.5	73.9	63.3	48.6	36.5
Mean Daily Temperature Range	5% DB	MDBR	17.7	19.1	21.0	22.5	23.7	22.7	23.7	22.2	23.9	21.8	18.5	15.3
		MCDBR	19.7	21.6	30.5	32.9	31.2	28.6	27.1	25.7	28.8	29.8	27.9	17.3
		MCWBR	16.7	16.4	19.2	18.5	15.0	13.0	12.6	11.9	15.3	18.3	19.4	15.0
	5% WB	MCDBR	18.2	21.1	26.0	28.1	25.9	23.0	23.1	21.2	23.5	25.5	25.3	15.5
		MCWBR	15.7	16.4	17.6	17.3	13.5	12.8	12.4	11.3	13.4	18.0	18.9	14.2
Clear-Sky Solar Irradiance	taub	0.261	0.308	0.327	0.363	0.396	0.403	0.396	0.393	0.362	0.325	0.292	0.267	
	taud	2.398	2.256	2.372	2.336	2.292	2.305	2.332	2.333	2.416	2.511	2.526	2.448	
	Ebn at Noon	279	278	289	285	278	275	276	272	274	272	266	266	
	Edh at Noon	23	33	34	38	41	40	39	37	32	25	21	21	
All-Sky Solar Radiation	RadAvg	533	876	1230	1437	1716	1892	1929	1657	1328	845	543	397	
	RadStd	63	81	137	139	131	137	95	109	103	93	53	40	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page